

Ameriforge

An AFG Holdings Company

Material Test Report

Heat Code: ED6

Ameriforge Ltd

13770 Industrial Rd. Houston, TX 77015

Sales: (713) 393-4200

ISO 9001:2015 Certified

NOV-R&M ENERGY SYS 500 CONROE PARK WEST DR CONROE, TX 77303	15278	PO: 4029723	Sales Order: 188181	Line: 2
		Item Code: 9999H61002-0031R	Qty Shipped: 1, 612	
		Item Desc: 2.00 150-900 HUB A350LF2 -17518204-001 NRM CDFRG		
		Mill Heat: C216029		
Spec: ASME SA350/SA350M(21) LF2 CL1 & NOV 61-0072 Rev. 01 (Hubs)				

Element	(%wt)	Ladle	Product	EPCRA	CAS#	Element	(%wt)	Ladle	Product	EPCRA	CAS#
C Carbon	0.19	✓	0.20			Cr Chromium	0.07	✓	0.07	Y	7440-47-3
Mn Manganese	1.12	✓	1.15	Y	7439-96-5	Mo Molybdenum	0.02	✓	0.01		
P Phosphorus	0.009	✓	0.008			V Vanadium	0.03	✓	0.03		
S Sulphur	0.01	✓	0.01			Cb Columbium	0.00	✓	0.01		
Si Silicon	0.25	✓	0.24			Al Aluminium	0.02	✓	0.02		
Cu Copper	0.15	✓	0.14	Y	7440-50-8	C.E.	0.42	✓			
Ni Nickel	0.06	✓	0.05	Y	7440-02-0						

Mechanical Testing		Heat Treat	
Test Lab	AFG	Norm. Temp. (F)	1675
Test Bar Size	SACRIFICIAL PIECE	Norm. (Hrs)	4
HBW	149 - 149	Norm. Cooling Type	AIR
Elg (%)	34		
RA (%)	63	Other	
Tensile Specimen Size (in)	.245	Supplier	STEEL DYNAMICS
Tensile (ksi)	82.4	Vacuum Degassed	Y
Yield (ksi)	52.3	Gr. Sz	8
Gauge Length (in)	1	Fully Killed	Y
Specimen Orientation	LONGITUDINAL	Made to Fine Grain Practice	YES
LCVN Charpy Temp. (F)	-50		
LCVN (ft lbs)	30 - 60 - 44		
LAvg 3 (ft-lbs)	44		
L Min (ft-lbs)	30		
4 Shear	10 - 40 - 30		
Lat. Exp.	.028 - .034 - .041		

Approved for Manufacture
by [Signature]
Date 11/1/22

The recording of false, fictitious, or fraudulent statement or entries on this document may be punishable as a felony under federal statute.

EPCRA Supplier Notification: This product may contain one or more toxic chemicals subject to the reporting requirements of Section 313 of the Emergency Planning and Community Right-to-Know Act (Title III of the Superfund Amendments and Reauthorization Act of 1980) and 40 C.F.R. Part 372. Potentially reportable chemicals are indicated with a checkmark in the "EPCRA" column and a Chemical Abstract Services (CAS) registry number is provided for each such chemical in addition to the percent by weight of the chemical present in this product. It is your responsibility alone to determine whether your facility is required to submit a Toxic Release Inventory Report under EPCRA Section 313.

Certification No.: 1610135
Certification Date: 10/10/2022
Issued By: Michelle Simien

This report is issued in compliance with the requirements of EN10204 3.1 / ISO 10474 3.1.b

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Sales: (713) 393-4200

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NOV-R&M ENERGY SYS	15278	PO: 4029723	Sales Order: 188181	Line: 2
500 CONROE PARK WEST DR		Item Code: 9999H61002-0031R	Qty Shipped: 1,612	
CONROE, TX 77303		Item Desc: 2.00 150-900 HUB A350LF2 -17518204-001 NRM CDFRG		
		MHI Heat: C216029		
Spec: ASME SA350/SA350M(21) LF2 CL1 & NOV 61-0072 Rev 01 (Hubs)				

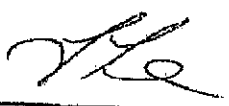
Product compliant with NACE MR0175/ISO 15156-2:2015, Annex A (A.2.1.2, A2.1.3).
Product compliant with NACE MR0103/ISO 17945:2015, Section 13.1.1 and 13.1.3.
TAG: H61002

- * No Weld repair performed.
- * Ladle Chemical Analysis results are reported from the raw material suppliers MTR.
- * Tensile taken from a round specimen. Measurement of elongation after fracture.
- * Yield strength was determined using the .2% offset method, unless specified.
- * All material supplied under this order is certified to be free of mercury contamination and no mercury bearing equipment was used in manufacturing, fabrication or testing.
- * When reported on MTR: Tensile Testing Per ASTM E8; Brinell Hardness Per ASTM E10; Charpy Testing Per ASTM E23
- * AF Global manufactures materials according to a certified Quality Management System conforming to ISO 9001:2015 and PED certified to 2014/68/EU, Annex 1, Section 4.3.
- Unless otherwise noted on PO, Dimensions of all standard pipe flanges and flange fittings (NPS 1/2 to NPS 24) per ASME B16.5-2020, large diameter steel flanges (NPS 26 to 60) per ASME B16.47-2020, Orifice flanges per ASME B16.36-2020.
- * AFGlobal uses the latest edition of ASTM E29 (Standard practice for using significant digits in test data to determine conformance with specifications), to claim conformance or nonconformance to a specification. When a accept/reject determination is required AFGlobal uses a test uncertainty ratio (TUR) of 1:1 or higher as acceptable.
- * DNV-GL PED Certificate No 10000425664-PA-ACCREDIA-USA Valid thru December 03, 2023.

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Certification Date: 10/10/2022
Issued By: Michelle Simian


Dusty Bannon - Quality Analyst

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13770 Industrial Rd. Houston, TX 77015

Sales: (713) 393-4200

NOV-R&M ENERGY SYS	15278	PO: 4033149	Sales Order: 191003	Line: 2
500 CONROE PARK WEST DR		Item Code: 9999H60002-0031R	Qty Shipped: 1,330	
CONROE, TX 77303		Item Desc: 02.00 150-900CAP A350LF2 17518043-001 NRM CDFRG		
		Mil Heat: C216029		
Spec: ASME SA105N/SA350 (17) ASME SA105N/SA350 (17) LF2 CL1 & NOV 61-0072 rev 1 (Caps)				

Element	(%wt)	Ladle	Product	EPCRA	CAS#	Element	(%wt)	Ladle	Product	EPCRA	CAS#
C Carbon	0.19	✓				Cr Chromium	0.07	✓			
Mn Manganese	1.12	✓		Y	7439-96-5	Mo Molybdenum	0.02	✓		Y	7440-47-3
P Phosphorous	0.009	✓				V Vanadium	0.034	✓			
S Sulphur	0.01	✓				Cb Columbium	0.00	✓			
Si Silicon	0.25	✓				Al Aluminum	0.02	✓			
Cu Copper	0.15	✓		Y	7440-50-8	C.E.	0.42	✓			
Ni Nickel	0.06	✓		Y	7440-02-0	CuNiCrMoV	0.33				

Mechanical Testing		Heat Treat	
Test Lab	AFG	Norm. Temp. (F)	1675
Test Bar Size	SACRIFICIAL PIECE	Norm. (Hrs)	1.30
HBW	146 - 146	Norm. Cooling Type	AIR
Elg (%)	32	Other	
RA (%)	61	Supplier	STEEL DYNAMICS
Tensile Specimen Size (in)	.247	Country of Melt	USA
Tensile (ksi)	81.2	Country of Forging	USA
Yield (ksi)	50.8	Vacuum Degassed	Y
Gauge Length (in)	1	Gr. Sz	8
Charpy Temp (F)	-50		
Charpy (ft lbs)	24 - 24 - 21		
Avg Energy 3 Specimens (ft-lbs)	24		
Min Energy 1 Specimens (ft-lbs)	21		
% Shear	10 - 10 - 10		
Lat. Exp. (in)	.021 - .020 - .017		

As by Date: *[Signature]* 4/1/23

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Certification No.: 1622149

Certification Date: 1/23/2023

Issued By: Michelle Gutierrez

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ISO 9001:2015 Certified

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NOV-R&M ENERGY SYS	15278	PO: 4033149	Sales Order: 191003	Line: 2
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CONROE, TX 77303		Item Desc: 02.00 150-900CAP A350LF2 17518043-001 NRM CDFRG		
		Min Heat: C216029		
Spec: ASME SA105N/SA350(17) ASME SA105N/SA350(17) LF2 CL1 & NOV 61-0072 rev 1 (Caps)				

Product compliant with NACE MR0175/ISO 15156-2:2015, Annex A (A.2.1.2, A2.1.3).
Product compliant with NACE MR0103/ISO 17943:2015, Section 13.1.1 and 13.1.3.

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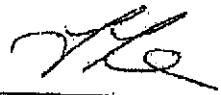
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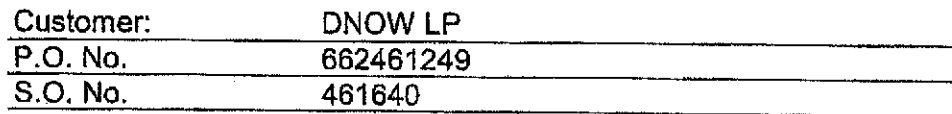
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Dusty Bannon - Quality Analyst

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